

Homework (Habanero)**Q1.** What is the vertex of the graph of $y = |x - 2| + 1$?

- (A) (2, 1)
 (B) (2, -1)
 (C) (1, 2)
 (D) (1, -2)
 (E) (-2, 1)

the y-intercept?

- (A) 1
 (B) 2
 (C) 3
 (D) 5
 (E) 6

Q2. A medicine dosage is modeled by $y = 2|x - 3|$. What is the minimum dosage?

- (A) 0
 (B) 2
 (C) 3
 (D) 4
 (E) 6

Q6. The graph of $y = |x + 1| - 3$ has a vertex at

- (A) (-1, -4)
 (B) (-1, -3)
 (C) (-1, -2)
 (D) (1, -3)
 (E) (1, 3)

Q3. The graph of $y = |x| - 2$ opens

- (A) upward
 (B) downward
 (C) to the left
 (D) to the right
 (E) it does not open

Q7. What is the equation of the graph of $y = |x - 2|$ after a horizontal shift of 2 units to the right?

- (A) $y = |x - 4|$
 (B) $y = |x + 2|$
 (C) $y = |x - 1|$
 (D) $y = |x|$
 (E) $y = |x + 4|$

Q4. What is the equation of the axis of symmetry of $y = |2x - 1|$?

- (A) $x = 0$
 (B) $x =$

Q8. The axis of symmetry of $y = |3x - 2|$ is

- (A) $x =$

rac23

- (A) $x =$

rac13

- (A) $x = 0$

- (B) $x = 2$

- (C) $x =$

rac32

rac12

- (B) $x = 1$

- (C) $x = 2$

- (D) $x =$

rac14

Q5. A cell growth model is given by $y = |x + 2| + 3$. What is**Open Ended Questions (FRQ)****Q9.** Graph the function $y = 2|x + 1| - 3$ and identify its vertex, axis of symmetry, and direction of opening.**Q10.** A patient's temperature is modeled by $y = |x - 4| + 2$, where x is the number of hours after medication is taken. If the patient's temperature is 6 degrees after 2 hours, does the model need to be adjusted? Explain.**Chef's Tips**

- Chef's Tips: For questions 9 and 10, check for completeness and accuracy of the work.
- Chef's Tips: Use a graphing calculator to check the graphs.
- Chef's Tips: Make sure to label the axes and provide a title for the graph.